Name : Tejas

USN : 2VX24UE059

Exp.No : 06

Hamiltonian Cycles Undirected Graph ¶

In [1]:

```python
n = int(input("Enter number of vertices in the graph: "))
print("Enter the adjacency matrix:")
graph = [list(map(int, input().split())) for _ in range(n)]

def find_hamiltonian_cycles(graph):
    n = len(graph); path = [0]; visited = [1] + [0]*(n-1); cycles = []
    def backtrack(v):
        if len(path) == n and graph[v][0]:
            cycles.append(path + [0]); return
        for u in range(n):
            if graph[v][u] and not visited[u]:
                visited[u] = 1; path.append(u)
                backtrack(u)
                path.pop(); visited[u] = 0
    backtrack(0); return cycles
print("==================================================")
cycles = find_hamiltonian_cycles(graph)
if cycles:
    print("Hamiltonian Cycles found:")
    for cycle in cycles:
        print(" -> ".join(map(str, cycle)))
else:
    print("No Hamiltonian Cycle found")
print("===============================================")
```

```
Enter number of vertices in the graph: 6
Enter the adjacency matrix:
 0 1 1 0 1 1
1 0 0 1 1 1
1 0 0 0 1 0
0 1 0 0 0 1
1 1 1 0 0 1
1 1 0 1 1 0
==================================================
Hamiltonian Cycles found:
0 -> 1 -> 3 -> 5 -> 4 -> 2 -> 0
0 -> 2 -> 4 -> 1 -> 3 -> 5 -> 0
0 -> 2 -> 4 -> 5 -> 3 -> 1 -> 0
0 -> 5 -> 3 -> 1 -> 4 -> 2 -> 0
===============================================
```